



**RAJARATA UNIVERSITY OF SRI LANKA
FACULTY OF APPLIED SCIENCES**

**Bachelor of Science in Information Technology
First Year - Semester II Examination – August / September 2024**

ICT 1308 –OPERATING SYSTEMS

Time: Three (03) hours

- This paper contains **FIVE (05)** questions on **THREE (03)** pages.
- Answer **ALL** questions.

1. a) There are two (02) main categories of computer software. Briefly explain them and give two (02) examples for each type. (05 marks)
 - b) Briefly explain two (02) functions of the operating systems. (04 marks)
 - c) What is a multiprogramming operating system? (03 marks)
 - d) Explain how a multiprogramming operating system improves system performance. (04 marks)
 - e) What are the typical applications of Real-Time Operating Systems (RTOS)? Choose two (02) applications and explain how an RTOS benefits them. (04 marks)
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2. a) Differentiate the terms “process” and “thread”. (04 marks)
 - b) Discuss the conditions under which a process might transition from the Waiting state to the Ready state. (04 marks)
 - c) Explain the differences between long-term, short-term, and medium-term schedulers. (05 marks)

- d) The following table indicates the CPU burst time and arrival time for six (06) processes P0, P1, P2, P3, P4, and P5.

Process	Arrival time(ms)	Burst time(ms)
P0	0	9
P1	1	3
P2	1	2
P3	1	4
P4	2	3
P5	3	2

Draw the Gantt charts for the following algorithms. (Assume the time quantum for RR is 3ms) (06 marks)

- a. Shortest Remaining Time First (SRTF)
 - b. Shortest Job First (SJF)
 - c. Round Robin (RR)
- ii. Find the average turnaround time, waiting time, and completion time for each of the above algorithms. (06 marks)
3. a) Describe the following terms related to operating systems. (06 marks)
- i. Process synchronization
 - ii. Race condition
 - iii. Convoy effect
- b) Explain what is a critical section and how the critical section problem is solved in operating systems. (04 marks)
- c) Discuss how semaphores are used in process synchronization. (04 marks)
- d) Explain how Peterson's solution works. (06 marks)

4. a) Differentiate between the terms "deadlock" and "starvation". (05 marks)
- b) What is a page fault in the context of demand paging? (05 marks)
- c) Explain the concept of internal and external fragmentation in the context of memory management. (05 marks)
- d) Briefly explain deadlock prevention techniques in your own terms. (05 marks)
5. a) What are the advantages and disadvantages of segmentation with reference to memory management? (*Note: You are required to provide two advantages and two disadvantages and briefly describe them.*) (04 marks)
- b) What is the difference between a "logical address" and a "physical address" in a paging system? (04 marks)
- c) Describe the different types of file access methods. (03 marks)
- d) Explain the differences between precise and imprecise interrupts. (04 marks)

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